



# Statistics of Financial Markets

## Seminar 8: Volatility estimators



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## Acronyms used in this material

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ ARCH-LM: ARCH Lagrange Multiplier (test)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ DM: Diebold–Mariano (test)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GK: Garman–Klass (volatility estimator)
- ▣ LB: Ljung–Box (test)
- ▣ LM: Lagrange Multiplier
- ▣ MSE: Mean Squared Error
- ▣ OHLC: Open, High, Low, Close
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ VIX: CBOE Volatility Index
- ▣ YZ: Yang–Zhang (volatility estimator)

## Today's Question and Route

- **Question:** how can we measure, from prices, a volatility that we never observe?
  - ▶ this seminar comes **before** Lecture 8: the section "What You Need for Today" gives every definition the tasks use
- Route
  - ▶ Part A: historical and EWMA volatility, range-based estimators, clustering tests and forecast losses, on paper
  - ▶ Part B: the S&P 500, BET, DAX, Bitcoin and BVB stocks, each task with an interpretation question
  - ▶ Part C: an open question for a project and an AI answer to audit
- Notebook for today: [open the seminar notebook in Google Colab](#); each task names its notebook section
- The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class

## Exercise Map

| Task    | Question                                                                 | Type             | Model  |
|---------|--------------------------------------------------------------------------|------------------|--------|
| A1, A2  | historical volatility, annualisation and its standard error              | Solved, Proposed | A1     |
| A3, A4  | EWMA step by step; half-life                                             | Solved, Proposed | A3     |
| A5, A6  | Parkinson, Garman–Klass, Rogers–Satchell and Yang–Zhang from OHLC prices | Solved, Proposed | A5     |
| A7, A8  | Ljung–Box on squared returns and the ARCH-LM test                        | Solved, Proposed | A7     |
| A9, A10 | MSE and QLIKE losses of volatility forecasts                             | Solved, Proposed | A9     |
| B1, B2  | rolling and EWMA volatility: S&P 500; BET, Bitcoin                       | Solved, Proposed | B1     |
| B3, B4  | range-based estimators: S&P 500; DAX, Bitcoin, TLV                       | Solved, Proposed | B3     |
| B5, B6  | clustering tests: S&P 500; BET, Bitcoin, TLV, SNP                        | Solved, Proposed | B5     |
| B7, B8  | forecast evaluation with the VIX; the best $\lambda$                     | Solved, Proposed | B7     |
| C1, C2  | which estimator for BVB stocks? what is wrong in an AI answer?           | Proposed         | B3, B7 |

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

## Data Used

| Series                                     | Source | Prices used           | Period    |
|--------------------------------------------|--------|-----------------------|-----------|
| S&P 500                                    | EODHD  | close; OHLC from 2008 | 1990–2026 |
| DAX                                        | EODHD  | close; OHLC from 2006 | 1990–2026 |
| BET                                        | EODHD  | close only            | 1997–2026 |
| Bitcoin                                    | EODHD  | OHLC, 7 days a week   | 2014–2026 |
| Banca Transilvania (TLV), OMV Petrom (SNP) | EODHD  | adjusted OHLC         | 2010–2026 |
| VIX                                        | EODHD  | close                 | 1990–2026 |

- Daily log returns in %:  $r_t = 100 (\ln P_t - \ln P_{t-1})$ ; weekends and repeated holiday closes are dropped (except Bitcoin); the last day is 18 September 2026
- Stocks: open, high and low multiplied by the same adjustment factor as the close (dividends, splits); TLV without 30–31 May 2016 ([Chapter 2](#))
- In the notebook: `returns('sp500')`, `ohlcv('btc')`; no account or key is needed

## What You Need for Today (1/5): Volatility and Its Precision

- ▣ **Volatility:** the standard deviation of returns; with  $n$  returns,  $\hat{\sigma} = \sqrt{\frac{1}{n-1} \sum_{t=1}^n (r_t - \bar{r})^2}$  (% per day)
  - ▶ zero-mean version for daily data:  $\hat{\sigma}^2 = \frac{1}{n} \sum_t r_t^2$
  - ▶ **historical volatility:** the same formula on a rolling window of the last  $n$  days ( $n = 21, 63, 252$ )
- ▣ **Annualisation:**  $\sigma_{\text{year}} = \sigma_{\text{day}} \sqrt{a}$ ,  $a$  = observations per year
  - ▶  $a \approx 252$  for stock markets,  $a = 365$  for Bitcoin; it assumes uncorrelated returns
- ▣ **Precision:** with  $n$  i.i.d. Normal returns,  $\text{SE}(\hat{\sigma})/\sigma \approx 1/\sqrt{2n}$ 
  - ▶ approximate 95% interval:  $\hat{\sigma} (1 \pm 1.96/\sqrt{2n})$

## What You Need for Today (2/5): EWMA

- **EWMA** (exponentially weighted moving average) variance, RiskMetrics J.P. Morgan/Reuters (1996):
  - ▶  $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$ , with the **decay factor**  $0 < \lambda < 1$ ;  $\lambda = 0.94$  for daily data
  - ▶  $\sigma_t^2$  uses returns up to day  $t - 1$ : it is the forecast for day  $t$
- Weights:  $\sigma_t^2 = (1 - \lambda) \sum_{j \geq 1} \lambda^{j-1} r_{t-j}^2$ 
  - ▶ **half-life**  $\ln 0.5 / \ln \lambda$ : the lag at which the weight halves
  - ▶ days that carry 99% of the weight:  $\ln 0.01 / \ln \lambda$
- A rolling window gives weight  $1/n$  to each of the last  $n$  days: when a big day leaves the window, the estimate drops abruptly (the **ghost effect**)

## What You Need for Today (3/5): Range-Based Estimators

- ▣ **OHLC** prices of day  $t$ : open  $O_t$ , high  $H_t$ , low  $L_t$ , close  $C_t$ ; logs in %:
  - ▶ overnight  $o_t = \ln(O_t/C_{t-1})$ ,  $u_t = \ln(H_t/O_t)$ ,  $d_t = \ln(L_t/O_t)$ ,  $c_t = \ln(C_t/O_t)$ ; close-to-close (CC)  
 $r_t = o_t + c_t$
- ▣ One-day variance estimates; on  $n$  days, take their mean
  - ▶ Parkinson [Parkinson \(1980\)](#):  $(u_t - d_t)^2 / (4 \ln 2)$ ; Garman–Klass [Garman & Klass \(1980\)](#):  
 $0.5(u_t - d_t)^2 - (2 \ln 2 - 1)c_t^2$
  - ▶ Rogers–Satchell [Rogers & Satchell \(1991\)](#):  $u_t(u_t - c_t) + d_t(d_t - c_t)$ , valid with a drift
- ▣ Yang–Zhang [Yang & Zhang \(2000\)](#) on  $n \geq 2$  days:  $\hat{\sigma}_O^2 + k\hat{\sigma}_C^2 + (1 - k)\hat{\sigma}_{RS}^2$ ,  $k = 0.34 / (1.34 + (n + 1)/(n - 1))$ 
  - ▶  $\hat{\sigma}_O^2$ ,  $\hat{\sigma}_C^2$ : sample variances of  $o_t$  and  $c_t$ ;  $\hat{\sigma}_{RS}^2$ : mean Rogers–Satchell term; Yang–Zhang is the only estimator that includes the night

## What You Need for Today (4/5): Volatility Clustering

- ▣ **Volatility clustering:** large moves are followed by large moves; returns are almost uncorrelated, but  $r_t^2$  and  $|r_t|$  are autocorrelated
  - ▶ sample **ACF** (autocorrelation function)  $\hat{\rho}_k$ ; 95% band for i.i.d. data:  $\pm 1.96/\sqrt{T}$
- ▣ **McLeod-Li test** [McLeod & Li \(1983\)](#): Ljung-Box [Ljung & Box \(1978\)](#) on  $r_t^2$ ,  
 $Q(m) = T(T+2) \sum_{k=1}^m \hat{\rho}_k^2 / (T-k) \sim \chi^2(m)$ 
  - ▶  $H_0$ : no clustering (constant conditional variance); reject if  $Q(m) > \chi_{0.95}^2(m)$
- ▣ **ARCH-LM test** [Engle \(1982\)](#): regress  $e_t^2$  ( $e_t = r_t - \bar{r}$ ) on a constant and  $e_{t-1}^2, \dots, e_{t-q}^2$ 
  - ▶  $LM = nR^2 \sim \chi^2(q)$ ,  $n$  = observations in the regression; 5% critical values:  $\chi_{0.95}^2(5) = 11.07$ ,  $\chi_{0.95}^2(10) = 18.31$

## What You Need for Today (5/5): Evaluating Forecasts

- A forecast  $h_t$  of tomorrow's variance is compared with a **proxy**, usually  $r_t^2$  (unbiased but noisy)
  - ▶ **MSE** (mean squared error): mean of  $(r_t^2 - h_t)^2$ ; **QLIKE**: mean of  $r_t^2/h_t + \ln h_t$ ; lower is better [Patton \(2011\)](#)
  - ▶ **DM** (Diebold–Mariano) test [Diebold & Mariano \(1995\)](#):  $t$  statistic of the mean loss difference of two forecasts;  $|t| > 1.96$ : they differ at 5%
- **VIX Cboe (2026)**: the expected volatility of the S&P 500 over the next 30 days, implied by option prices, in annualised %
  - ▶ as a forecast of the daily variance:  $h_t = \text{VIX}_{t-1}^2/a$
- Forecasts must use only information available at the close of day  $t - 1$

## A1: Historical Volatility from Five Days [Solved]

### Task

- ▣ Daily returns (%) of a stock: 1.2, -0.8, 0.5, -1.5, 0.6.
- ▣ 1. Compute the mean and the sample standard deviation (divide by  $n - 1$ ).
- ▣ 2. Annualise the standard deviation with  $a = 252$ .
- ▣ 3. Compute the zero-mean volatility  $\sqrt{\frac{1}{n} \sum r_t^2}$ .
- ▣ 4. Compute  $1/\sqrt{2n}$  and an approximate 95% interval for the annual volatility.
- ▣ Report: the mean, two daily volatilities, one annual volatility, one interval and one sentence.

### Solution

- ▣ 1. Mean =  $0/5 = 0$ ; squared deviations 1.44, 0.64, 0.25, 2.25, 0.36, sum 4.94;  $s^2 = 4.94/4 = 1.235$ ,  $s = 1.111\%$
- ▣ 2.  $1.111 \times \sqrt{252} = 17.6\%$
- ▣ 3.  $\sqrt{4.94/5} = 0.994\%$ : with a zero mean, only the divisor changes
- ▣ 4.  $1/\sqrt{10} = 31.6\%$ ;  $17.6 \times (1 \pm 1.96 \times 0.316)$ : [6.7%; 28.6%]
- ▣ Five days say almost nothing: the annual volatility could be anywhere between 6.7% and 28.6%.

## A2: Bitcoin over Six Days [Proposed]

Solution discussed in class

### Task

- ▣ Daily Bitcoin returns (%): 3.1, -4.2, 1.0, 5.5, -2.4, -0.9. Model: A1.
- ▣ 1. Compute the mean and the sample standard deviation.
- ▣ 2. Annualise it with  $a = 365$  and, wrongly, with  $a = 252$ .
- ▣ 3. Compute the approximate 95% interval for the annual volatility ( $a = 365$ ).
- ▣ Report: two annual volatilities, one interval and one sentence.

## A3: EWMA Step by Step [Solved]

### Task

- ▣  $\lambda = 0.94$ ; today's EWMA variance is  $\sigma_1^2 = 1.00$  ( $\%^2$ ); the next three returns are  $r_1 = -2.0\%$ ,  $r_2 = 0.5\%$ ,  $r_3 = 3.0\%$ .
- ▣ 1. Compute  $\sigma_2^2$ ,  $\sigma_3^2$  and  $\sigma_4^2$ .
- ▣ 2. Annualise  $\sigma_4$  with  $a = 252$ .
- ▣ 3. Compute the half-life and the weight of  $r_3^2$  in  $\sigma_4^2$ .
- ▣ Report: three variances, one annual volatility, the half-life and one sentence.

### Solution

- ▣ 1.  $\sigma_2^2 = 0.94 \times 1.00 + 0.06 \times 4.00 = 1.1800$
- ▣  $\sigma_3^2 = 0.94 \times 1.1800 + 0.06 \times 0.25 = 1.1242$ ;  
 $\sigma_4^2 = 0.94 \times 1.1242 + 0.06 \times 9 = 1.5967$
- ▣ 2.  $\sigma_4 = 1.264\%$ ;  $1.264 \times \sqrt{252} = 20.1\%$
- ▣ 3.  $\ln 0.5 / \ln 0.94 = 11.2$  days; the newest squared return gets weight  $1 - \lambda = 0.06$
- ▣ One 3% day raises the variance by 42%; a quiet day lowers it by only 6%.

## A4: A Slower EWMA [Proposed]

Solution discussed in class

### Task

- ▣ The same data as A3, with  $\lambda = 0.97$  (the RiskMetrics monthly value). Model: A3.
- ▣ 1. Compute  $\sigma_2^2$ ,  $\sigma_3^2$  and  $\sigma_4^2$  and annualise  $\sigma_4$ .
- ▣ 2. Compute the half-life and the number of days that carry 99% of the weight.
- ▣ 3. Compare with A3: which  $\lambda$  reacts faster to the 3% day?
- ▣ Report: three variances, one annual volatility, two numbers of days and one sentence.

## A5: Four Estimators from One Day [Solved]

### Task

- ▣ Previous close 100.0; open 101.0, high 103.5, low 99.8, close 102.2.
- ▣ 1. Compute  $o$ ,  $u$ ,  $d$ ,  $c$  (in %) and  $r = o + c$ .
- ▣ 2. Compute the one-day CC, Parkinson, Garman–Klass and Rogers–Satchell variances.
- ▣ 3. Annualise each with  $a = 252$ , as if every day looked like this one.
- ▣ 4. Say which estimators see the overnight move  $o$ .
- ▣ Report: five log returns, four variances, four annual volatilities and one sentence.

### Solution

- ▣ 1.  $o = 100 \ln(101/100) = 0.995$ ,  $u = 2.445$ ,  $d = -1.195$ ,  $c = 1.181$ ,  $r = 2.176$
- ▣ 2. CC  $r^2 = 4.736$ ; Parkinson  $3.640^2/2.773 = 4.780$ ; GK  $0.5 \times 3.640^2 - 0.386 \times 1.181^2 = 6.087$ ; RS 5.931
- ▣ 3.  $\sqrt{252 \hat{\sigma}^2}$ : CC 34.5%, Parkinson 34.7%, GK 39.2%, RS 38.7%
- ▣ 4. Only CC (and Yang–Zhang, on several days) include  $o$ ; Parkinson, GK and RS see only the session.

## A6: Three Days and the Yang–Zhang Estimator [Proposed]

Solution discussed in class

### Task

| Day | Open | High | Low  | Close |
|-----|------|------|------|-------|
| 1   | 50.2 | 51.0 | 49.6 | 50.8  |
| 2   | 50.5 | 51.9 | 50.3 | 51.6  |
| 3   | 51.9 | 52.4 | 50.9 | 51.1  |

- Previous close: 50.0. Model: A5.
- 1. Compute  $o$ ,  $u$ ,  $d$ ,  $c$  for each day.
- 2. Compute the Parkinson and Rogers–Satchell terms of each day and their means.
- 3. Compute  $k$ ,  $\hat{\sigma}_O^2$ ,  $\hat{\sigma}_C^2$  and the Yang–Zhang variance for  $n = 3$ .
- 4. Compare with the CC sample variance and annualise all three with  $a = 252$ .
- Report: a small table, three variances, three annual volatilities and one sentence.

## A7: Two Tests for Clustering [Solved]

### Task

- ▣  $T = 1000$  daily returns; the ACF of the squared returns at lags 1–5: 0.20, 0.15, 0.12, 0.10, 0.08; the ARCH-LM regression with  $q = 5$  lags on  $n = 995$  observations has  $R^2 = 0.08$ .
- ▣ 1. Compute the Ljung–Box statistic  $Q(5)$  of the squared returns.
- ▣ 2. Compute  $LM = nR^2$ .
- ▣ 3. Compare both with  $\chi^2_{0.95}(5) = 11.07$  and decide.
- ▣ Report: two statistics, two decisions and one sentence.

### Solution

- ▣ 1. Terms  $1000 \cdot 1002 \hat{\rho}_k^2 / (1000 - k)$ : 40.12, 22.59, 14.47, 10.06, 6.45;  $Q(5) = 93.69$
- ▣ 2.  $LM = 995 \times 0.08 = 79.60$
- ▣ 3. Both far above 11.07 (both  $p < 0.001$ ): reject  $H_0$ ; the variance is predictable from the recent past.

## A8: When the Two Tests Disagree [Proposed]

Solution discussed in class

### Task

- ▣  $T = 500$ ; ACF of the squared returns at lags 1–5: 0.12, 0.08, 0.06, 0.05, 0.04; ARCH-LM with  $q = 5$  on  $n = 495$  observations:  $R^2 = 0.018$ . Model: A7.
- ▣ 1. Compute  $Q(5)$  and decide at 5%.
- ▣ 2. Compute LM and decide at 5%.
- ▣ 3. Give one reason why the two tests can disagree.
- ▣ Report: two statistics, two decisions and one sentence.

## A9: MSE and QLIKE by Hand [Solved]

### Task

- ▣ Over four days the proxy  $r_t^2$  is 1.0, 4.0, 0.25, 2.25 (%<sup>2</sup>). Forecast A is flat at 1.5; forecast B is 1.0, 2.5, 0.8, 1.8.
- ▣ 1. Compute the MSE of each forecast.
- ▣ 2. Compute the QLIKE terms  $r_t^2/h_t + \ln h_t$  and their mean for each forecast.
- ▣ 3. Say which forecast is better under each loss.
- ▣ Report: two MSE values, two QLIKE values and one sentence.

### Solution

- ▣ 1. A:  $(1 - 1.5)^2 = 0.2500, 6.2500, 1.5625, 0.5625$ , MSE 2.156; B: 0.0000, 2.2500, 0.3025, 0.2025, MSE 0.689
- ▣ 2. A:  $1/1.5 + \ln 1.5 = 1.072, 3.072, 0.572, 1.905$ , mean 1.655; B: 1.000, 2.516, 0.089, 1.838, mean 1.361
- ▣ 3. B is better under both losses: it follows the changes of the variance; a flat forecast misses them.

## A10: Under- and Over-Prediction [Proposed]

Solution discussed in class

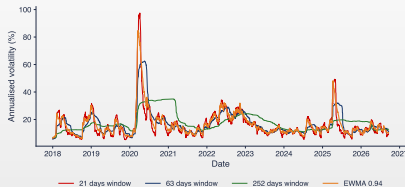
### Task

- ▣ The true variance and the proxy are both 1. Two forecasts:  $h = 0.5$  (half) and  $h = 2$  (double). Model: A9.
- ▣ 1. Compute the MSE of each forecast.
- ▣ 2. Compute the QLIKE of each forecast.
- ▣ 3. Say which loss punishes under-prediction more.
- ▣ Report: four numbers and one sentence on why this matters for VaR.

## B1: Historical and EWMA Volatility of the S&P 500 [Solved]

- ▣ **Question:** how different are the 21-, 63- and 252-day and EWMA volatilities of the S&P 500?
- ▣ **Inputs:** S&P 500 closes since 1990, daily log returns in %
- ▣ **Tasks**
  1. Compute the rolling 21-, 63- and 252-day volatility and the EWMA volatility ( $\lambda = 0.94$ ), annualised with the actual frequency.
  2. Draw the four series since 2018.
  3. Report the four values at the end of the sample and their maxima in 2020, with dates.
  4. Find the day of June 2020 when the 63-day volatility falls most, and report the change of EWMA on that day.
  5. Interpretation: which estimate would you report to a risk committee on that day?
- ▣ **Report:** the chart, twelve numbers and two sentences
- ▣ **Notebook:** section B1

## B1: Solution [Solved]



- $a = 252$ ; whole sample 18.0%; on 18 September 2026: 21 days 9.6%, 63 days 11.1%, 252 days 12.9%, EWMA 10.4%
- 2020 maxima: 97.5% (6 April 2020), 62.6% (20 May 2020), 34.7% (28 December 2020), EWMA 84.8% (24 March 2020)
- On 15 June 2020 the 63-day volatility fell from 50.4% to 43.0% (the crash day of 16 March left the window); EWMA moved from 35.6% to 34.6%
- Interpretation: EWMA; the drop of the 63-day estimate is the ghost effect, not news; the 252-day value still reflects March

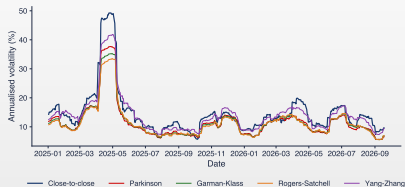
## B2: BET and Bitcoin [Proposed]

- **Question:** how volatile are the BET and Bitcoin compared with the S&P 500, today and at the end of 2017? Model: B1.
- **Inputs:** BET since 1997, Bitcoin since 2014, S&P 500 since 1990, daily log returns in %
- **Tasks**
  1. Compute the number of observations per year and the whole-sample annual volatility of each series.
  2. Annualise Bitcoin also with 252 and give the error in %.
  3. Report the 252-day volatility at the end of 2017 and at the end of the sample, and the maximum of the 63-day volatility with its year.
  4. Interpretation: has Bitcoin become an asset with equity-like volatility?
- **Report:** one table and two sentences
- **Notebook:** section B2

## B3: Range-Based Estimators for the S&P 500 [Solved]

- ▣ **Question:** do range-based estimators give the same volatility as closing prices for the S&P 500?
- ▣ **Inputs:** S&P 500 OHLC since 2008 ( $T = 4\,707$  days)
- ▣ **Tasks**
  1. Compute  $o_t$ ,  $u_t$ ,  $d_t$ ,  $c_t$  and the five estimators on the whole sample, annualised.
  2. Decompose  $\text{Var}(r_t) = \text{Var}(o_t) + \text{Var}(c_t) + 2 \text{Cov}(o_t, c_t)$ .
  3. Draw the rolling 21-day estimates since 2025.
  4. Interpretation: why are Parkinson, Garman–Klass and Rogers–Satchell below close-to-close?
- ▣ **Report:** the chart, five volatilities, a decomposition and two sentences
- ▣ **Notebook:** section B3

## B3: Solution [Solved]



- Annual volatility: CC 19.9%, Parkinson 15.4%, GK 14.6%, RS 14.4%, YZ 16.3%
- Daily variances ( $\%^2$ ):  $1.568 = 0.176 + 1.194 + 0.197$ ; the night is 13% of  $\text{Var}(o) + \text{Var}(c)$ ;  $\text{corr}(o, c) = 0.22$
- Interpretation: these estimators see only the trading session, so they miss the overnight variance and the covariance term; YZ adds the night but not the covariance

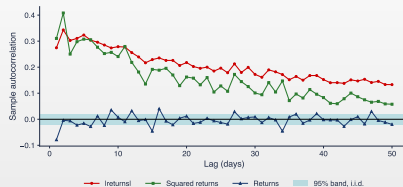
## B4: DAX, Bitcoin and TLV [Proposed]

- **Question:** for which of the DAX, Bitcoin and Banca Transilvania do range-based estimators agree with close-to-close?  
Model: B3.
- **Inputs:** DAX OHLC since 2006, Bitcoin since 2014, TLV adjusted OHLC since 2010
- **Tasks**
  1. Compute the five annualised estimators for each asset.
  2. Compute the share of the night and  $\text{corr}(o_t, c_t)$ .
  3. Rank the assets by the ratio Parkinson / CC.
  4. Interpretation: for which asset could you replace close-to-close by Parkinson?
- **Report:** one table and two sentences
- **Notebook:** section B4

## B5: Clustering in the S&P 500 [Solved]

- ▣ **Question:** are the size and the sign of S&P 500 returns equally predictable?
- ▣ **Inputs:** S&P 500 closes since 1990, daily log returns in %
- ▣ **Tasks**
  1. Draw the ACF of  $r_t$ ,  $r_t^2$  and  $|r_t|$  for lags 1–50 with the i.i.d. 95% band.
  2. Report  $\hat{\rho}_1, \dots, \hat{\rho}_5$  of  $r_t$  and of  $r_t^2$ .
  3. Compute the Ljung–Box  $Q(10)$  of  $r_t$ ,  $r_t^2$  and  $|r_t|$ , and the ARCH-LM(5) statistic with its  $R^2$ .
  4. Interpretation: does volatility clustering contradict weak-form market efficiency?
- ▣ **Report:** the chart, ten autocorrelations, four statistics and two sentences
- ▣ **Notebook:** section B5

## B5: Solution [Solved]



- $\hat{\rho}_{1..5}(r)$ :  $-0.079, -0.004, -0.006, -0.022, -0.016$ ;  $\hat{\rho}_{1..5}(r^2)$ :  $0.311, 0.408, 0.250, 0.298, 0.308$ ;  $T = 9\,240$
- $Q(10)$ :  $r\,90.8$  ( $p < 0.001$ ),  $r^2\,8\,015$ ,  $|r|\,8\,323$  (critical value  $18.31$ );  $\text{ARCH-LM}(5) = 9\,235 \times 0.241 = 2\,227$  (critical value  $11.07$ )
- Interpretation: no; the size is predictable, the sign hardly; efficiency is about predictable returns, not about predictable risk ([Chapter 7](#))

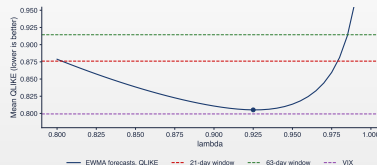
## B6: Four More Series [Proposed]

- **Question:** does volatility clustering in the BET, Bitcoin, Banca Transilvania and OMV Petrom depend on the order of the days? Model: B5.
- **Inputs:** BET since 1997, Bitcoin since 2014, TLV and SNP since 2010
- **Tasks**
  1. Compute  $\hat{\rho}_1(r^2)$ ,  $Q(10)$  of  $r_t^2$ , ARCH-LM(5) and the excess kurtosis of each series.
  2. Shuffle the returns at random (seed 2026) and compute  $Q(10)$  of  $r_t^2$  and ARCH-LM(5) again.
  3. Interpretation: why does shuffling remove the clustering but not the heavy tails?
- **Report:** one table and two sentences
- **Notebook:** section B6

## B7: Which Forecast Is Best? [Solved]

- ▣ **Question:** which simple forecast of tomorrow's S&P 500 variance has the smallest loss since 2010?
- ▣ **Inputs:** S&P 500 returns since 1990, VIX closes; forecasts for 4 202 days from 2010
- ▣ **Tasks**
  1. Build next-day forecasts: rolling means of  $r_t^2$  on 21, 63 and 252 days, EWMA with  $\lambda = 0.94$  and  $0.97$ , and  $VIX^2/a$ .
  2. Compute MSE and QLIKE with the proxy  $r_t^2$ .
  3. Run the Diebold–Mariano test for EWMA 0.94 against the 21-day window and for the VIX against EWMA 0.94.
  4. Draw the QLIKE of EWMA for  $\lambda$  from 0.80 to 0.995.
  5. Interpretation: is the VIX a better forecast of tomorrow's variance than EWMA?
- ▣ **Report:** a table of twelve numbers, two tests, the chart and two sentences
- ▣ **Notebook:** section B7

## B7: Solution [Solved]



|       | 21    | 63    | 252   | EWMA 0.94 | EWMA 0.97 | VIX   |
|-------|-------|-------|-------|-----------|-----------|-------|
| MSE   | 18.99 | 20.77 | 21.43 | 17.77     | 19.02     | 16.70 |
| QLIKE | 0.876 | 0.914 | 1.099 | 0.808     | 0.844     | 0.799 |

- DM: EWMA 0.94 against 21 days  $t = -4.50$  ( $p < 0.001$ ); VIX against EWMA 0.94  $t = -0.32$  ( $p = 0.751$ ); best  $\lambda = 0.925$
- Interpretation: no; the VIX has slightly lower losses, but the difference is not significant; both beat the rolling windows

## B8: The Best $\lambda$ for the BET and Bitcoin [Proposed]

- **Question:** is the RiskMetrics  $\lambda = 0.94$  a good choice for the BET and for Bitcoin? Model: B7.
- **Inputs:** BET since 1997 and Bitcoin since 2014; forecasts from 2010 (BET) and from 2016 (Bitcoin)
- **Tasks**
  1. Compute the QLIKE of next-day EWMA forecasts for  $\lambda$  from 0.80 to 0.995 and find the best  $\lambda$ .
  2. Report the QLIKE at the best  $\lambda$ , at 0.94 and at 0.97, and the half-life of the best  $\lambda$ .
  3. Run the Diebold–Mariano test of the best  $\lambda$  against 0.94.
  4. Interpretation: would you use  $\lambda = 0.94$  for both series?
- **Report:** one table and two sentences
- **Notebook:** section B8

## C1: Which Estimator for BVB Stocks? [Proposed]

- ▣ **Question:** do range-based estimators improve volatility forecasts for thinly traded stocks of the Bucharest Stock Exchange?
- ▣ **Inputs:** adjusted OHLC of TLV, SNP, BRD and TGN since 2010; S&P 500 since 2008 as a benchmark; models: B3, B7
- ▣ **Tasks**
  1. Compute the share of trading days with an unchanged close, and the ratios Parkinson / CC and YZ / CC of the whole-sample volatility.
  2. Compute the QLIKE (proxy  $r_t^2$ ) of next-day forecasts by the 21-day CC, Parkinson and YZ variances, from 2010.
  3. Compare the four stocks with the S&P 500.
  4. List two reasons why the best estimator may differ across stocks.
  5. Interpretation: what data would you need to explain the differences?
- ▣ **Report:** a table and a plan for a project
- ▣ **Notebook:** section C1

## C2: Audit an AI Answer [Proposed]

- A student asked an AI assistant to summarise the volatility of the S&P 500 and of Bitcoin. The answer:
- (a) Since 2008 the Parkinson volatility of the S&P 500 is 15.4% and the close-to-close volatility 19.9%; Parkinson is about five times more efficient, so the true volatility is 15.4%.
- (b) The Ljung-Box  $Q(10)$  of S&P 500 returns is 90.8 ( $p < 0.001$ ), so the S&P 500 shows volatility clustering.
- (c) EWMA with  $\lambda = 0.94$  has a half-life of about 94 days.
- (d) The daily standard deviation of Bitcoin is 3.49%, so its annual volatility is  $3.49 \times \sqrt{252} = 55.3\%$ .
- (e) Since 1990 the VIX was above the realised volatility of the next 21 days on about 86% of days.
- Tasks
  - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
  - ▶ 2. Report: a list of five verdicts with one line of justification each.

## What You Should Take from Today

- ▣ Annualise with the actual frequency: 252 for stock markets, 365 for Bitcoin
- ▣ Short windows react fast, long windows smooth the estimate; EWMA reacts without ghost effects
- ▣ Range-based estimators are precise but see only the session: use Yang–Zhang when the market closes at night
- ▣ Clustering is tested on  $r_t^2$  (Ljung–Box, ARCH-LM), not on  $r_t$
- ▣ Compare forecasts with QLIKE and test the difference; an AI answer is a draft to check

## After the Seminar

- Lecture 8 develops each topic of today: historical and EWMA volatility, range-based and realised volatility, clustering, the VIX, forecast evaluation
- Try the [Proposed] tasks in the notebook; the solutions are discussed in class
- C1 can grow into a team project: all BET stocks, liquidity groups, Diebold–Mariano tests
- Reading: [Franke, Härdle & Hafner \(2019\)](#), Ch. 13; exercises in [Borak, Härdle & López-Cabrera \(2013\)](#), Ch. 13; [Tsay \(2010\)](#), Ch. 3.1

## References

- ▣ Borak, S., Härdle, W.K., López-Cabrera, B. (2013). *Statistics of Financial Markets: Exercises and Solutions*, 2nd ed. Springer.
- ▣ Cboe Global Markets (2026). *Cboe Volatility Index (VIX)*. Product page and methodology.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Garman, M.B., Klass, M.J. (1980). On the estimation of security price volatilities from historical data. *Journal of Business*, 53(1), 67–78.
- ▣ J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document*, 4th ed. New York: Morgan Guaranty Trust Company.
- ▣ Ljung, G.M., Box, G.E.P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303.
- ▣ McLeod, A.I., Li, W.K. (1983). Diagnostic checking ARMA time series models using squared-residual autocorrelations. *Journal of Time Series Analysis*, 4(4), 269–273.
- ▣ Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.
- ▣ Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Rogers, L.C.G., Satchell, S.E. (1991). Estimating variance from high, low and closing prices. *Annals of Applied Probability*, 1(4), 504–512.
- ▣ Tsay, R.S. (2010). *Analysis of Financial Time Series*, 3rd ed. Wiley.
- ▣ Yang, D., Zhang, Q. (2000). Drift-independent volatility estimation based on high, low, open, and close prices. *Journal of Business*, 73(3), 477–492.